S1. Model Assumptions. Strengths and Limitations

Model Assumptions

Markov Model Structure

- The simulation follows a single cohort of 100,000 individuals entering the UK at one time [1]
- The model runs in monthly cycles over a 55-year lifetime horizon (660 monthly cycles) [2]
- In the base-case model, transmission beyond the cohort is not modelled explicitly. Downstream secondary case prevention is explored separately in Scenario 2 using a simplified transmission proxy rather than a dynamic transmission model.
- Emigration is not explicitly accounted for in the simulation
- Screening takes place at the beginning of the simulation at t=0

Population & Epidemiology

- Cohort entry age is in the 25-29 years band [3], [4]
- Rate of progression from latent to active TB is two-phase: 0.000912/month (years 0–5 and 0.000333/month (years 5–55) [4]
- Background mortality is age-dependent, with 11 5-year-long bands (ages 25–29 to 75–79) [5]
- For Active TB cases, disease-specific mortality is incorporated alongside background mortality
- Multidrug-resistant tuberculosis (MDR-TB) is out of scope of the simulation; only 2.4% of people with positive cultures had a pattern of resistance in 2023 in England [6]
- The diagnostic data on the screening pathways are used in creating the initial distribution of migrant health states in the Markov model [1]

Screening and Treatment

- The standard treatment for LTBI is 3HR (3 months), £64/month [3], [7]
- The standard treatment for Active TB is 2HRZE/4HR (6 months), £770/month [3], [8]

- LTBI individuals who haven’t completed treatment re-join LatentDiagnosed category without further re-treatment
- Individuals with false-positive testing results rejoin the Uninfected category

Costs
- The productivity losses from disease and societal costs are out of scope of the simulation [2]
- All costs are expressed in 2023/24 GBP
- Diagnostic costs are accounted for once at t=0
- Undiagnosed LTBI cases have no associated costs
- Lost-to-follow-up LTBI cases have no associated costs
- Undiagnosed Active TB cases have no associated costs

Health Utilities
- The associated utility for Uninfected and LTBI individuals (including those undergoing treatment) is 0.92 [9]
- The associated utility for Active TB (excluding individuals undergoing treatment) is 0.68 [11], [10]
- The associated utility for Active TB individuals undergoing treatment is 0.81 [11]
- The associated utility for individuals who have completed Active TB treatment (ActiveCompleted) is the same as for Uninfected individuals
- The associated utility for dead individuals is 0

Discounting
- The discounting rate for costs and utilities is 3.5% per year [2]
- Monthly discounting rate is calculated as $(1.035)^{(1/12)} - 1 \approx$ 0.287%/month

PSA
- The model runs 1,000 Monte Carlo simulations
- A fixed random seed is used for reproducibility of the analysis
- Probabilities and utilities are beta-distributed [12], [13]
- Costs are gamma-distributed [12], [13]

- Initial conditions are not varied in PSA
- The primary willingness-to-pay (WTP) threshold was £25,000/QALY, with £35,000/QALY explored as an upper NICE threshold in probabilistic sensitivity analysis and interpretation of cost-effectiveness. [2]

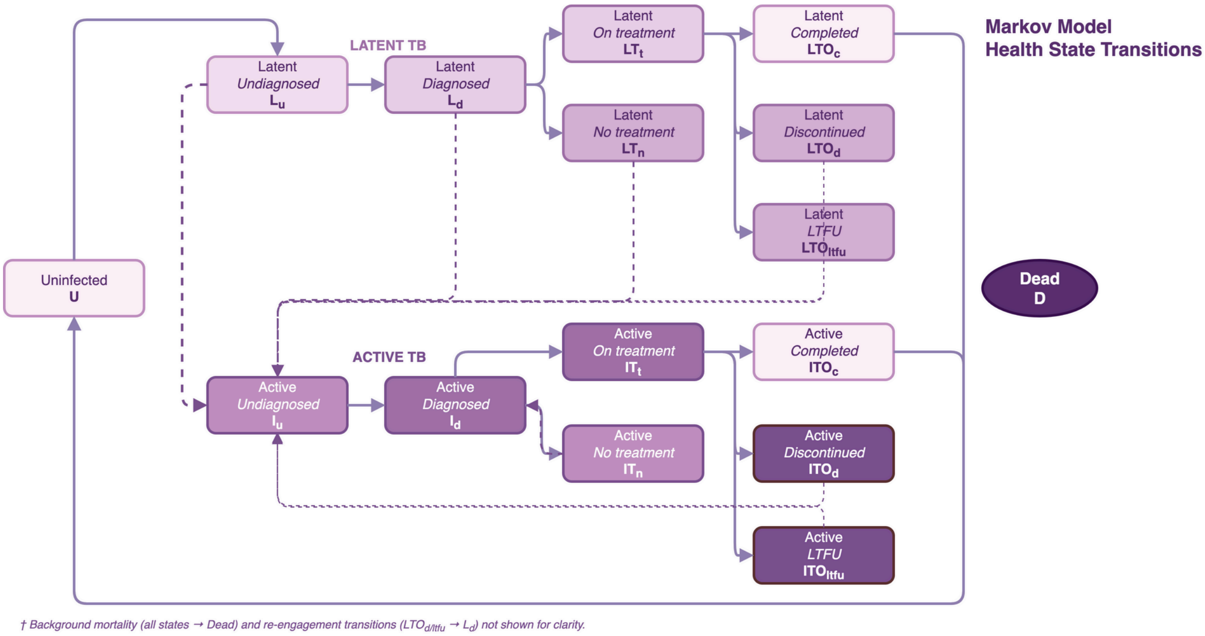

Figure 1. Markov model health states and transitions. The model comprises 16 health states across three disease domains: uninfected, latent TB infection (LTBI), and active TB disease. Arrows represent permitted monthly transitions; not all transitions are shown for clarity. A dead state (absorbing) is reached via TB-related or background mortality from any state (not shown). Initial state occupancy is determined by the upstream screening decision tree and varies by strategy. Background mortality is applied from all states using age-varying rates (ONS National Life Tables 2020–22). LTBI = latent tuberculosis infection; LTFU = lost to follow-up; ONS = Office for National Statistics; QALY = quality-adjusted life year; TB = tuberculosis.

Strengths and Limitations

The core strengths of this analysis lie in its two-fold methodology. The hybrid structure (screening decision tree + Markov model) allows analysis and comparison of screening strategies with complex screening patient cascades and strategies that differ in their ability to detect both active TB and LTBI. Rigorous uncertainty analysis through both PSA and DSA quantifies the robustness of the results.

The design of Markov models allows for tracking long-term changes in individuals' health states, which is well-suited for the natural history of tuberculosis.

The assumed homogeneity of the cohort is important for the model's clarity and straightforwardness, but it also limits its ability to capture insights into how migrants' individual characteristics, such as age, HIV status, and country of origin, affect model results. The possible ways to address the model's limitations in the future are further discussed in the Future Work subsection.

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